INTERNSHIP REPORT & PROJECT REVIEW-1

Project: AI Resume Screening System

Internship Period: [18-05-2026] to [17-07-2026]

[B . Chennakesavulu] – [24691A2838] – III CST Sec A

Review Date: [21-10-2026]

Internship Mentor: [Mr . M . Shankar, Asst. Professor of CST **]**

About Industry / Organization Details

- **Organization name: [**MOTHERSON automotive technologies**]**
- **What it does: [**Motherson Technology Services Limited is a public, non-government technology and industrial solutions company within the Motherson Group, offering digital, cloud, AI, and IT services while also being legally classified under security-dealing activities as per MCA filings.]
- **Address:** [4th Floor, Wing C, Tower 3, Casagrand Ecotech, 475-A, Wipro Street, Sholinganallur, Chennai – 600119, Tamil Nadu]
- **Project domain:** Web Development and Resume Screening
- **Project type:** Flask-based web application for initial resume review

Internship Certificate



motherson

24th Aug 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Chennakesavulu**, from **Madanapalle Institute of Technology & Science** has done his Internship Training at Motherson Automotive Technologies & Engineering, **Unit - II**, starting from 18th May 2026 to 17th July 2026.

During the Internship Period his conduct was good.

Wishing him every success.

For Motherson Automotive Technologies and Engineering
(A Division of Samvardhana Motherson International Limited)

Authorised Signatory

My Internship

Project Description

- Project: AI Resume Screening System
- Purpose: Compare PDF resumes with a job description
- Technologies: Python, Flask, HTML, CSS, JavaScript, SQLite, pypdf

Things Learned

- Building Flask routes and web pages
- Uploading and reading PDF files
- Finding skill keywords with rules
- Calculating match scores
- Saving records and exporting CSV

Specific Tasks

Week 1	Reviewed the resume-screening workflow and project requirements.
Week 2	Created job-title and job-description inputs with PDF upload validation.
Week 3	Extracted readable PDF text and identified recognized skill keywords.
Week 4	Compared required skills and calculated matched and missing skills.
Week 5	Saved candidate details in SQLite and tested dashboard and CSV export.

The Project Proposal

AI Resume Screening System

Title of Our Project

AI Resume Screening System

PDF Resume Skill Extraction and Keyword-Based Job Matching

Objective of the Project

1. **Upload:** Accept a job title, job description, and multiple PDF resumes.
2. **Extract:** Read text from supported, text-based PDF resumes.
3. **Match:** Identify recognized skills and find matched or missing skills.
4. **Display:** Show a keyword-based score and save candidate results.

Abstract

- The AI Resume Screening System is a web-based application that helps recruiters review resumes easily. Recruiters enter the job title and job description and upload PDF resumes.
- The system uses Python Flask to process the uploaded resumes. The pypdf library extracts text, and predefined keywords are used to identify candidate skills.
- The system compares the candidate's skills with the required job skills and calculates a matching percentage. Candidate details, scores, matched skills, and missing skills are stored in an SQLite database.
- Recruiters can view candidate records and export the results as a CSV file. The system uses keyword matching, not a trained AI model, so recruiters must manually review every resume before making hiring decisions.

Existing System and Its Disadvantages

Existing approaches

- Manual reading of each resume
- Comparing resumes with job requirements by hand
- Tracking candidates in separate files or spreadsheets
- Searching for skills one resume at a time

Disadvantages

- Takes more time for many resumes
- Repeated checking of similar skills
- Important skills may be overlooked
- Candidate records can be difficult to organize
- Manual comparison may be inconsistent

Proposed System with Advantages

Proposed system

- Upload up to 20 PDF resumes per batch
- Extract text using the pypdf library
- Identify recognized skills with keyword rules
- Show matched skills, missing skills, and score
- Store candidates in SQLite and export CSV

Advantages

- Organizes screening in one interface
- Reduces repeated manual comparison
- Makes skill gaps easier to review
- Keeps candidate records together
- Supports quick CSV-based reporting

Hardware & Software Requirements

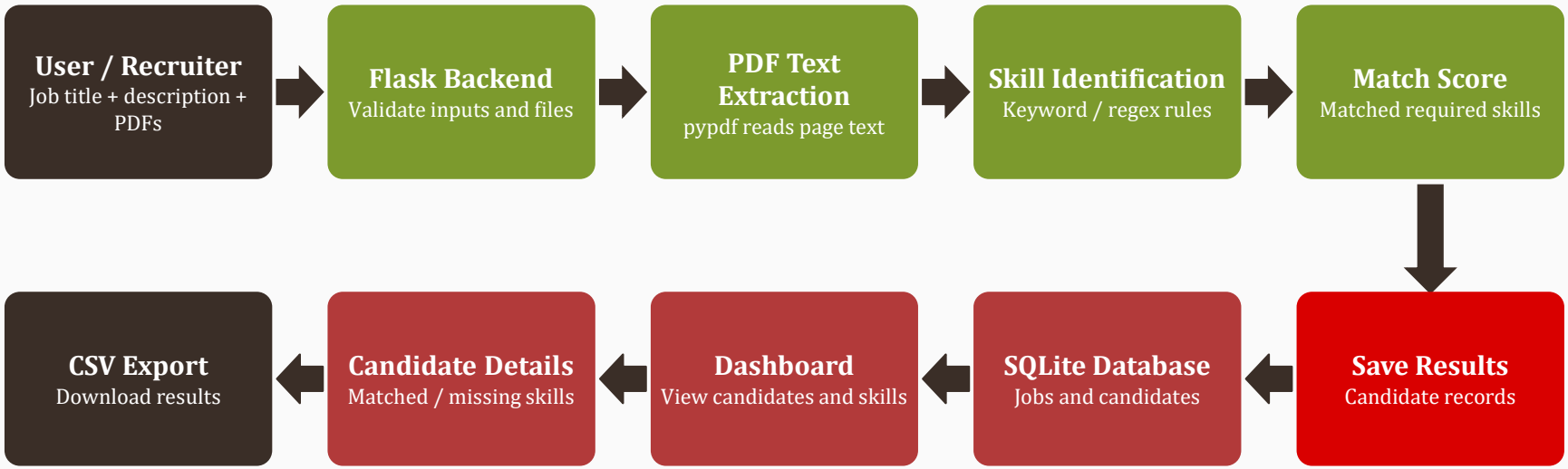
Hardware

- Laptop or desktop computer
- Keyboard, monitor, and storage
- Internet connection for setup
- Recommended: 4 GB RAM or more

Software

- OS: Windows or Linux
- Language: Python 3.10+
- Backend: Flask
- Frontend: HTML, CSS, JavaScript
- Database: SQLite
- PDF library: pypdf
- Code Editor: Visual Studio Code
- Web Browser: Google Chrome or Microsoft Edge

System Architecture



Flow: User input → Flask validation → PDF text extraction → skill matching → score calculation → SQLite storage → dashboard and CSV export.

Methodologies Used for Implementation (1/2)

Input and preprocessing

- Enter the job title and job description

Upload PDF resumes (maximum 20 per batch)

Validate PDF file type, signature, and size

Extract readable text from PDF pages with pypdf

Skill identification

- Search text for recognized skills using keyword rules

Use regular expressions for selected skill names and aliases

Compare resume skills with skills in the job description

Record matched and missing required skills

Methodologies Used for Implementation (2/2)

Scoring and output

- Score = matched required skills ÷ recognized required skills × 100

Store job and candidate information in SQLite

Display skills, scores, and candidate details

Export candidate records to CSV

Limitations and future work

- This version uses keyword rules, not a trained ML model

Scanned PDFs need OCR; text-based PDFs are supported

The score is not a hiring recommendation

Future work: DOCX, OCR, better skill extraction, access control

References

1. Python Software Foundation. Python Documentation. https://docs.python.org/3/
2. Pallets Projects. Flask Documentation. https://flask.palletsprojects.com/
3. SQLite. SQLite Documentation. https://www.sqlite.org/docs.html
4. pypdf contributors. pypdf Documentation. https://pypdf.readthedocs.io/
5. MDN Web Docs. HTML, CSS and JavaScript. https://developer.mozilla.org/
6. W3C. CSV on the Web: A Primer. https://www.w3.org/TR/tabular-data-primer/

Conclusion

- The system organizes the initial screening of PDF resumes.
- It extracts resume text and identifies candidate skills.
- It compares skills with job requirements and displays a keyword-based score.
- Candidate details are stored in SQLite, with options to view records and export CSV files.
- It reduces repetitive manual work but does not use a trained AI model or make hiring decisions.
- Future Enhancement: Add DOCX support, OCR for scanned resumes, better privacy controls, and advanced skill extraction.

THANK YOU